Unit 1: Overview of Electronic Components & Signals

1. Passive Components: Resistors, Capacitors, and Inductors

> **Definition: Passive components are electronic devices that cannot introduce net energy into a circuit and cannot amplify a signal. They can only attenuate, store, or dissipate energy.**

Any electrical or electronic circuit is fundamentally built using basic building blocks. Passive components form the most basic physical foundation of these circuits. They do not require any external power source to operate.

- **Resistor (R):** A component designed to oppose or limit the flow of electric current. It dissipates energy in the form of heat. Governed by Ohm's Law (V = IR). Its unit is the Ohm (Ω).
- **Capacitor (C):** A component that stores electrical energy in the form of an electrostatic field. It consists of two conducting plates separated by an insulating material (dielectric). It blocks DC but allows AC to pass. Its unit is the Farad (F). Formula: Q = CV.
- **Inductor (L):** A coil of wire that stores electrical energy in the form of a magnetic field when current passes through it. It opposes any sudden change in current. It allows DC to pass but blocks high-frequency AC. Its unit is the Henry (H). Formula: V = L(di/dt).

Applications

- **Resistors:** Current limiting, voltage division, biasing active components.
- **Capacitors:** Filtering noise, smoothing power supplies, tuning radio frequencies, coupling/decoupling signals.
- **Inductors:** Chokes in power supplies, transformers, tuning circuits (oscillators).

```
SYMBOLS OF PASSIVE COMPONENTS:

 Resistor:     ---/\/\/\/\---

 Capacitor:    ---| |---

 Inductor:     ---UUUU---
```

👉 **Hinglish Explanation:** Passive components wo hote hain jo signal ko amplify (bada) nahi kar sakte aur inko kaam karne ke liye alag se battery/power nahi chahiye. **Resistor** current ko rokta hai (speed breaker). **Capacitor** charge ko temporary store karta hai (electric field mein) aur DC ko rok leta hai. **Inductor** ek coil hai jo magnetic field mein energy store karta hai aur current ke achanak change hone ka virodh karta hai.

2. Active Components: Diodes & Transistors (BJT)

> **Definition: Active components are devices that can inject power into a circuit, control the flow of current, and amplify signals. They require an external power source to operate.**

Active components are made of semiconductor materials (like Silicon and Germanium) and are the core of modern electronics, enabling switching, logic, and amplification.

1. Diodes:

A diode is a two-terminal (Anode and Cathode) semiconductor device formed by joining a P-type and an N-type material (PN Junction). It acts as a one-way valve for electric current.

- **Forward Bias:** When Anode is positive and Cathode is negative, it conducts current easily.
- **Reverse Bias:** When connected in reverse, it blocks the current flow.
- **Applications:** Rectifiers (converting AC to DC), voltage regulators (Zener diode), light emission (LEDs).

2. Transistors (BJT - Bipolar Junction Transistor):

A BJT is a three-terminal device (Emitter, Base, and Collector) that uses a small input current at the base to control a much larger current flowing between the collector and emitter.

- **Types:** NPN (arrow points Out from base) and PNP (arrow points In towards base).
- **Operation:** It operates in three regions: Cut-off (acts as open switch), Saturation (acts as closed switch), and Active region (acts as an amplifier).
- **Applications:** Signal amplification (audio/radio), digital logic gates, electronic switches.

```
DIODE SYMBOL:
      Anode (P)      Cathode (N)
        ----|>|----

NPN TRANSISTOR SYMBOL:
           C
          /
    B ---|
          \
           v  E  (Arrow outwards)
```

> 👉 **Hinglish Explanation:** Active components external power lekar signal ko amplify karte hain. **Diode** ek one-way rasta hai, ye current ko sirf ek direction mein jaane deta hai (AC ko DC banane ke kaam aata hai). **Transistor** electronics ka dil hai. Isme 3 taange hoti hain (Base, Emitter, Collector). Base par thoda sa current dene se baaki dono ke beech bada current flow hota hai. Ye as a Switch (on/off) aur as an Amplifier (aawaz badi karna) use hota hai.

3. FET, MOS, and CMOS & Their Applications

Definition: Field Effect Transistors (FET) are unipolar active semiconductor devices where current is controlled by an electric field (voltage). MOS and CMOS are advanced, highly efficient families of FET technology.

While BJTs are current-controlled devices, FETs are **Voltage-Controlled Devices**. This gives them very high input impedance, meaning they draw almost zero current from the input source.

1. FET (Field Effect Transistor):

It has three terminals: **Gate (G), Drain (D), and Source (S)**. Voltage applied at the Gate controls the flow of current between the Drain and Source.

2. MOSFET (Metal Oxide Semiconductor FET):

An advanced FET where the Gate is completely insulated from the main conducting channel by a thin layer of Silicon Dioxide (SiO_2).

- **Advantages:** Extremely high input impedance, fast switching speed, and can be scaled down to microscopic sizes.
- **Types:** NMOS (uses electrons, faster) and PMOS (uses holes).

3. CMOS (Complementary MOS):

CMOS technology combines both NMOS and PMOS transistors in a single circuit in a complementary way. When the NMOS is ON, the PMOS is OFF, and vice versa.

- **Why is CMOS so important?** Because one transistor is always OFF, a CMOS circuit consumes virtually **ZERO static power**. It only consumes power when switching states.

Applications of MOS & CMOS:

CMOS is the dominant technology in the world today. It is used to manufacture Microprocessors (Intel/AMD CPUs), Microcontrollers, RAM memory chips, Image sensors in digital cameras, and mobile phone processors.

> 👉 **Hinglish Explanation:** BJT current se control hota hai, jabki **FET** Voltage se control hota hai. Isme Gate, Drain aur Source hote hain. **MOSFET** me Gate ke neeche ek oxide ki layer hoti hai jisse wo aur accha kaam karta hai. Sabse zaroori **CMOS** hai, jo NMOS aur PMOS ko milakar banta hai. Iska sabse bada fayda ye hai ki ye battery/power bahut kam khata hai. Aajkal ke saare computer processors (CPU) aur mobile chips CMOS technology se hi bante hain.

4. Signals: DC/AC, Voltage/Current, Periodic/Non-Periodic

Definition: In electrical engineering, a signal is any physical quantity (voltage or current) that varies with time and carries information.

1. Voltage and Current Signals:

- **Voltage Signal:** Represents the potential difference across two points varying over time.
- **Current Signal:** Represents the flow of electric charge through a conductor varying over time.

2. AC vs. DC Signals:

- **Direct Current (DC):** A signal whose magnitude remains constant and direction never changes with time. Example: Battery

output.

- **Alternating Current (AC):** A signal whose magnitude changes continuously and direction reverses periodically. Example: Household power supply.

3. Periodic vs. Non-Periodic Signals:

- **Periodic Signal:** A signal that repeats its exact pattern after a fixed interval of time (called Time Period, T). Mathematically, $f(t) = f(t + T)$. Example: A pure sine wave, clock pulses.
- **Non-Periodic (Aperiodic) Signal:** A signal that does not repeat its pattern over time. Example: Human speech, internet data signals, random noise.

4. Different Types of Signal Waveforms:

- **Sine Wave:** The most fundamental, smooth, naturally occurring AC waveform.
- **Square Wave:** Alternates instantly between two distinct voltage levels (High and Low). Used in digital electronics.
- **Triangular Wave:** Rises and falls linearly at a constant rate.
- **Sawtooth Wave:** Rises linearly and drops instantly (or vice versa). Used in CRT displays and synthesizers.

```
WAVEFORMS:

 SINE WAVE:       SQUARE WAVE:       TRIANGULAR WAVE:
   /\                ___               /\
  /  \              |   |             /  \
 /    \             |   |            /    \
-------+----   -------+---+---   -------+----+---
        \  /              |   |               \  /
         \/               |___|                \/
```

> 👉 **Hinglish Explanation:** Signal ek aisi voltage ya current hai jo time ke sath badalti hai aur information le jati hai. **DC** signal time ke sath change nahi hota (ekdum seedhi line). **AC** signal upar-neeche hota hai. Agar signal apna design ek fix time ke baad wapas repeat kare, toh use **Periodic** kehte hain (jaise sine wave). Agar repeat na kare, toh **Non-periodic** kehte hain (jaise hamari aawaz). Digital computers me Square wave use hoti hai.

5. Signal Parameters: Peak, Average, and RMS Values

> **Definition: Since AC signals constantly change their value over time, specific mathematical parameters (Peak, Average, and RMS) are used to measure and define their magnitude for practical circuit calculations.**

Consider a standard AC voltage sine wave given by the equation: $v(t) = V_m \sin(\omega t)$.

1. Peak Value (V_m or I_m):

The maximum displacement or maximum value reached by the alternating quantity during one complete cycle. Also known as the amplitude.
Peak-to-Peak Value (V_{p-p}): The total voltage from the positive peak to the negative peak ($2 \times V_m$).

2. Average Value (V_{avg}):

The arithmetic mean of all instantaneous values of an alternating quantity over one complete cycle.
For a pure, symmetrical sine wave, the positive half and negative half cancel each other out, making the **Average Value over a full cycle = 0**.
Therefore, the average value is usually calculated over a *half-cycle*. For a sine wave: **$V_{avg} = \frac{2 V_m}{\pi} \approx 0.637 V_m$**.

3. RMS Value (Root Mean Square) / Effective Value (V_{rms}):

This is the most important parameter. The RMS value of an AC current is defined as that equivalent DC current which produces the **exact same amount of heat** in a given resistor as produced by the AC current in the same time.
For a sine wave, the relationship is: **$V_{rms} = \frac{V_m}{\sqrt{2}} \approx 0.707 V_m$**.
Note: When we say our home power supply is "230 Volts", it refers to the **RMS value**. The actual peak voltage is $230 \times \sqrt{2} \approx 325$ Volts.

> 👉 **Hinglish Explanation:** AC voltage har pal badalta hai, toh hum uski strength kaise napenge? **Peak value** sabse high point hai. **Average value** pure cycle ka 0 hota hai, isliye half cycle ka nikalte hain (0.637 times of peak). Sabse zaroori **RMS value** hai. RMS wo DC voltage hai jo AC ke barabar garmi (heat) paida karega. Hamare ghar mein jo 230V aata hai, wo RMS value hi hota hai. RMS = Peak / $\sqrt{2}$.

6. Electrical Sources: Ideal/Non-ideal & Independent/Dependent

> **Definition: An electrical source is a device that supplies electrical energy (voltage or current) to a circuit. They are classified based on their internal characteristics and their dependency on other circuit variables.**

1. Ideal vs. Non-Ideal Sources:

- **Ideal Voltage Source:** Provides a constant voltage output regardless of the current drawn by the load. Its internal resistance is exactly **Zero**. (Practically impossible).
- **Non-Ideal (Practical) Voltage Source:** Has a small internal resistance connected in *series*. As load current increases, the output voltage drops slightly due to this internal resistance ($V = V_s - I R_{in}$). Example: A real car battery.
- **Ideal Current Source:** Provides a constant current output regardless of the voltage across it. Its internal resistance is **Infinite**.
- **Non-Ideal (Practical) Current Source:** Has a high (but not infinite) internal resistance connected in *parallel* with the source.

2. Independent vs. Dependent Sources:

- **Independent Sources:** Their output voltage or current is fixed and does not depend on any other voltage or current in the circuit. They are represented by a **Circle** in circuit diagrams. Example: A standard 9V battery.
- **Dependent (Controlled) Sources:** Their output value depends directly on a voltage or current existing somewhere else in the same circuit. They are represented by a **Diamond** shape in circuit diagrams. They are mainly used to model the behavior of active components like transistors and amplifiers.

Types of Dependent Sources:

1. **VCVS** (Voltage Controlled Voltage Source): Output voltage depends on input voltage.
2. **CCVS** (Current Controlled Voltage Source): Output voltage depends on input current.

3. **VCCS** (Voltage Controlled Current Source): Output current depends on input voltage.
4. **CCCS** (Current Controlled Current Source): Output current depends on input current.

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SYMBOLS:

 Independent V-Source    Dependent V-Source
      .---.                    /\
     /  +  \                  / +\
    |   V   |                <  V >
     \  -  /                  \ -/
      '---'                    \/
     (Circle)                (Diamond)
```

👉 **Hinglish Explanation:** Source do tarah ke hote hain. **Ideal** source matlab 100% perfect, jo kabhi voltage drop nahi karta (iske andar resistance 0 hoti hai). Par real duniya me battery aisi nahi hoti, unme thodi internal resistance hoti hai, inhe **Non-ideal** kehte hain. **Independent** source apni marzi se chalta hai (Circle symbol). Lekin **Dependent** source (Diamond symbol) circuit ke kisi aur hisse ki voltage/current par nirbhar karta hai (jaise Transistor ki working me hota hai).

🔁 Quick Revision Notes (Exam Saver)

- **Passive vs Active:** Passive (R, L, C) store/dissipate energy. Active (Diodes, BJT, FET) can amplify and control power.
- **CMOS:** Combines NMOS and PMOS. Dominates modern chips due to practically zero static power consumption.
- **Signals:** Periodic repeats after time T (Sine, Square). DC is constant, AC alternates.
- **AC Parameters:** RMS is the effective heating value ($V_m / \sqrt{2}$). Average of full sine cycle is 0.
- **Sources:** Ideal voltage source has 0 internal resistance. Ideal current source has infinite internal resistance.
- **Dependent Sources:** Represented by a Diamond shape. Value depends on another circuit variable (VCVS, CCVS, VCCS, CCCS).

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